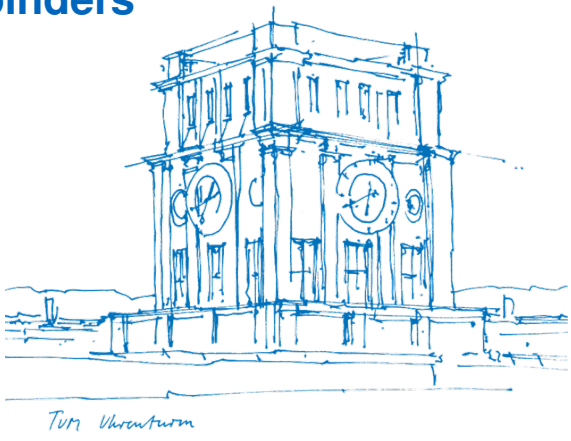
TUM School of Computation, Information and Technology
Department of Computer Science
Technical University of Munich

Property-Conditioned Generative Design of Minibinders

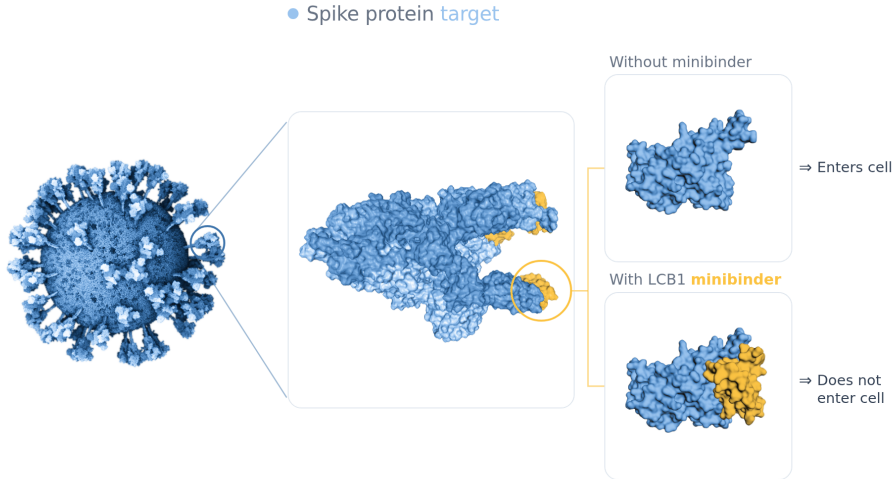
Martin Biroščák

Clinical Application Project

Advisor Leon Hetzel

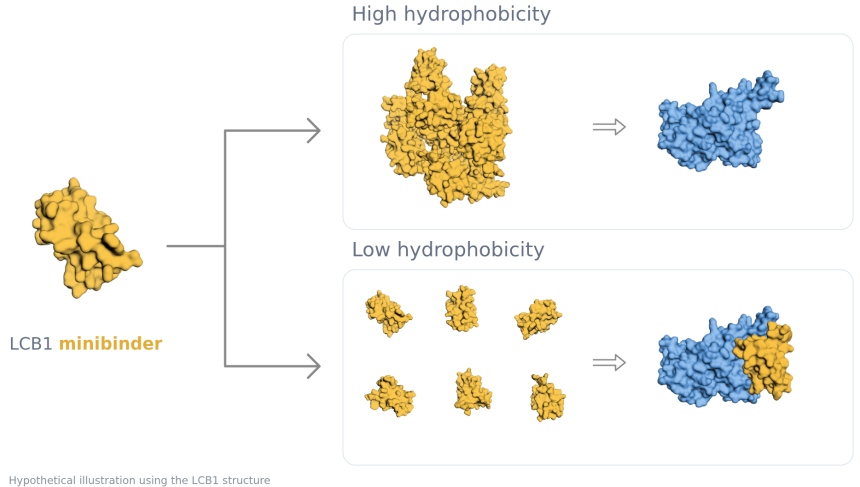


What are minibinders?



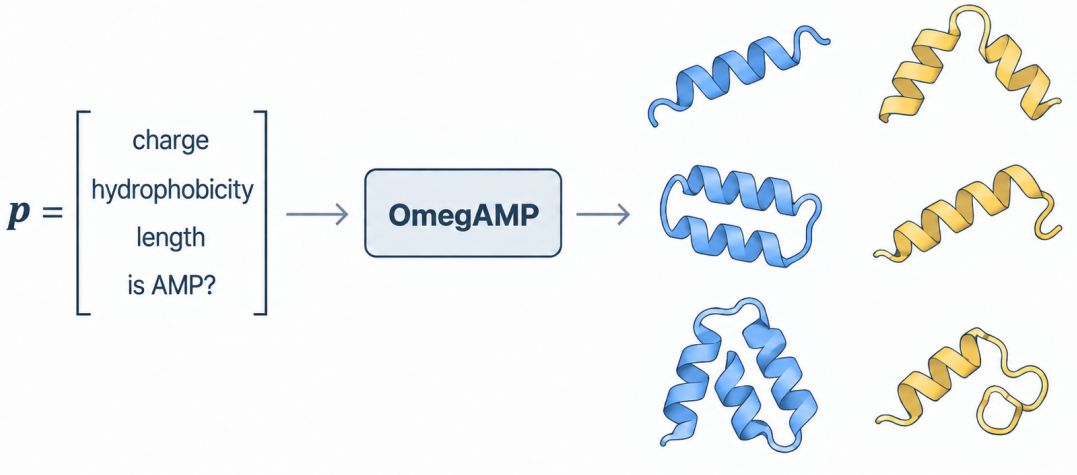
Spike/LCB1 structures from Cao et al. (2020).
Science **370**, 426–431. PDB 7JZL/7JZU. Leftmost virus rendering adapted from Nanographics (recoloured; CC BY 4.0)

Minibinders

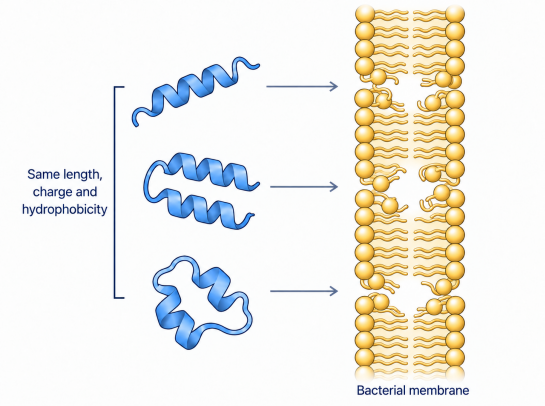


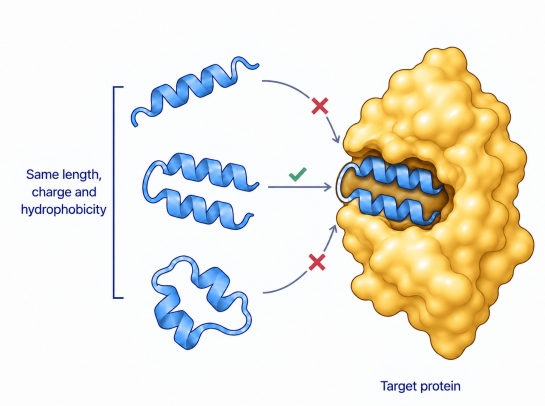
LCB1 structure from Cao et al. (2020). *Science* **370**, 426–431. PDB 7JZU

OmegAMP

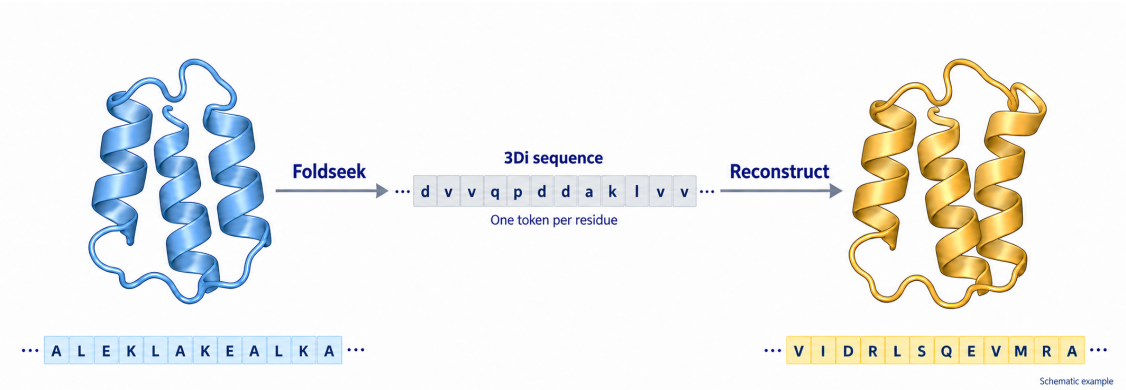


Challenge

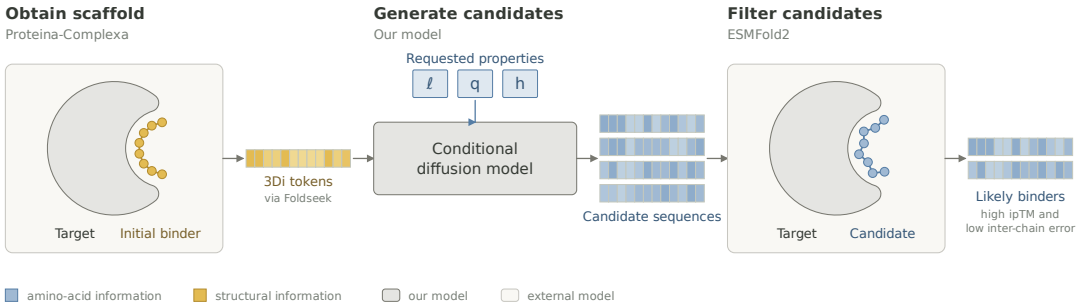

Same length,
charge and
hydrophobicity
Bacterial membrane


Same length,
charge and
hydrophobicity
Target protein

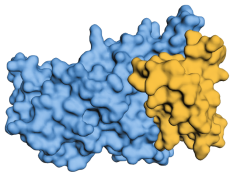
How do we specify shape?



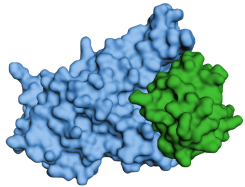
Generative pipeline



Does it work?

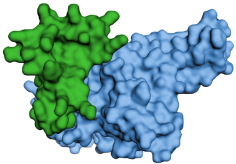


Original

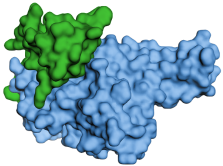


Candidate 16

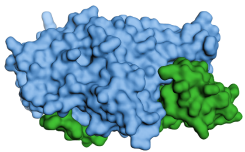
Charge	Hydrophobicity	Binder pLDDT	ipTM
-0.265	-0.0884	0.668	0.570



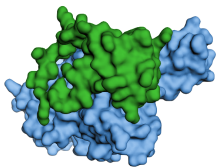
Candidate 163



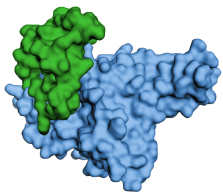
Candidate 168



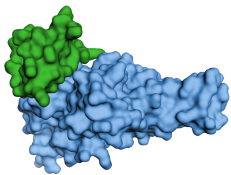
Candidate 312



Candidate 350



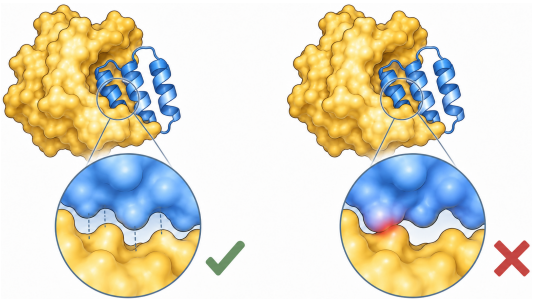
Candidate 397



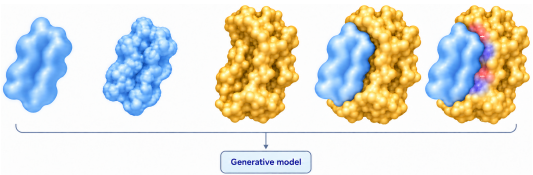
Candidate 467

Discussion

We underspecified the binder and binding is highly dependent on the details.



Future direction: explore other representations



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Department of Computer Science
Technical University of Munich

Thank you

